

STOCHASTIC METHODS IN WATER RESOURCES

Unit 4: Time series analysis

Lecture 12: Univariate, linear stochastic models

Luis Alejandro Morales, Ph.D.

Universidad Nacional de Colombia
Department of Civil and Agriculture Engineering

October 27, 2025

Introduction

- ▶ Here we will deal with **univariate linear models** used to represent **stochastic process**. The main purpose of this models is to generate likely **future sequences of data** for **design** and **planning**.
- ▶ **Generated data sequences** are particularly useful in determining the **reliabilities** of maintaining **specific conditions** from complex systems.
- ▶ In general, the models are formulated so that the current value of a variable is the **weighted sum** of **past values** and **random numbers** which represent unknown effects.
- ▶ Models such **autoregressive (AR)**, **moving average (MA)** and **Box-Jenkins autoregressive moving average (ARMA)** and **autoregressive integrated moving-averaged (ARIMA)** models are explained.
- ▶ Initially, it is assumed that **trend** and **periodicity** are removed prior to model application. Then, **seasonal models** are explained.
- ▶ **Stochastic models** are based on the following assumptions:
 1. It is assumed that the **sample data** is representative of the population.
 2. The condition of **stationary** (and **ergodicity**) must be invoked. However, **non-stationary** models can be formulated.
 3. Time series must be represent **natural processes** so any enforced effects on the series due for instance to a dam construction must be accounted.
 4. It is expected that the models maintain the statistical characteristics of the processes. This means that the synthetic generated series must be realistic.
- ▶ In practice, all these assumptions are seldom met. Nevertheless, the best possible model must be established based on the available data.

Linear autoregressive models

- ▶ In a **linear autoregressive model**, the **current value** is equal to the **weighted sum** of a preassigned number of **past values** and a variate that is **random**.
- ▶ Note that **linear** means that the past values have exponent equal to 1.
- ▶ These models are a first approximation to many **natural processes**. For instance, **streamflow** is constituted by the addition of time dependent processes such **groundwater flow**, **surface run-off**, **catchment retention**, etc.
- ▶ These models are applied to the **stochastic component** ξ_t of **stochastic process**, X_t represented by $X_t = T_t + P_t + \xi_t$, where T_t is the **trend** and P_t is the **periodic component**.
- ▶ It is assumed that the **trend** and the **seasonality** are removed from the original series and the remaining, ξ_t , is treated as a **random variable**.
- ▶ The models say that the value of ξ_t at time t is equal to the weighted sum of p past values at times $t - 1, t - 2, \dots, t - p$ and a random number η_t , where η_t $t = 0, \pm 1, \pm 2, \dots$, values are mutually independent and identically distributed.

Linear autoregressive models

pth-order model

The **pht-order model linear autoregressive mode AR(p)** takes the form

$$\xi_t = \phi_{p,1}\xi_{t-1} + \phi_{p,2}\xi_{t-2} + \phi_{p,3}\xi_{t-3} + \cdots + \phi_{p,p}\xi_{t-p} + \eta_t = \sum_{i=1}^p \phi_{p,i} \xi_{t-i} + \eta_t \quad (1)$$

where $\phi_{p,i}$, $i = 1, 2, 3, \dots, p$ are the **autoregressive parameters** or **weights**. The properties of ξ_t and η_t are $E[\xi_t] = E[\eta_t] = 0$ and $Var[\xi_t] = E[\xi_t^2] = \sigma_\xi^2$,

$Var[\eta_t] = E[\eta_t^2] = \sigma_\eta^2$, $\rho_k = E[\xi_t \xi_{t-k}] / \sigma_\xi^2$ and, for $k = 1, 2, 3, \dots$,

$E[\eta_t \eta_{t-k}] = E[\eta_t \xi_{t-k}] = 0$. This last equality means that current random number is independent of past values of the process.

It is assumed without loss of generality that the variance σ_ξ^2 of the **stochastic component** is equal to 1. This means that, in application, the numbers generated through such equation must be multiply by the **standard deviation** of the variable modelled and the **mean** should then be added to each number.

Linear autoregressive models

Estimation of autoregressive parameters

If the equation above is multiply by ξ_{t-1} and **expectations** are taken

$$\begin{aligned} E[\xi_t \xi_{t-1}] &= \phi_{p,1} E[\xi_{t-1} \xi_{t-1}] + \phi_{p,2} E[\xi_{t-2} \xi_{t-1}] + \phi_{p,3} E[\xi_{t-3} \xi_{t-1}] \\ &+ \cdots + \phi_{p,p} E[\xi_{t-p} \xi_{t-1}] + E[\eta_t \xi_{t-1}] \end{aligned}$$

Because $E[\eta_t \xi_{t-1}] = 0$ and on account of the other properties given above, it follows that

$$\rho_1 = \phi_{p,1} + \phi_{p,2} \rho_1 + \phi_{p,3} \rho_2 + \cdots + \phi_{p,p} \rho_{p-1}$$

Furthermore, if the model equation is multiplied by $\xi_{t-2}, \xi_{t-3}, \dots, \xi_{t-p}$ in turn and expectations are taken after each multiplication, p relations called the **Yule-Walker** equations are obtained. In matrix form, these equations are

$$\begin{bmatrix} \rho_1 \\ \rho_2 \\ \rho_3 \\ \vdots \\ \vdots \\ \rho_p \end{bmatrix} = \begin{bmatrix} 1 & \rho_1 & \rho_2 & \cdots & \rho_{p-1} \\ \rho_1 & 1 & \rho_1 & \cdots & \rho_{p-2} \\ \rho_2 & \rho_1 & 1 & \cdots & \rho_{p-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho_{p-1} & \rho_{p-2} & \rho_{p-3} & \cdots & 1 \end{bmatrix} \begin{bmatrix} \phi_{p,1} \\ \phi_{p,2} \\ \phi_{p,3} \\ \vdots \\ \vdots \\ \phi_{p,p} \end{bmatrix} \quad (2)$$

In a compact way

$$\rho_p = \mathbf{P}_p \phi_p$$

A necessary condition for **stationarity** is that the autocorrelation matrix $\mathbf{P}_p$ is **positive definite**, that is, the determinant and all its principal minors are zero or positive.

Therefore

$$\phi_p = \mathbf{P}_p^{-1} \rho_p$$

Linear autoregressive models

Estimation of autoregressive parameters

In order to solve the equation above, **serial correlation coefficients** $r_1, r_2, \dots, r_p$ are substituted in $\mathbf{P}_p$ and ρ_p , and hence the **Yule-Walker** estimates of the **autoregressive parameters** $\phi_{p,1}, \phi_{p,2}, \dots, \phi_{p,p}$ are obtained. For a normal **linear process**, this gives the **maximum likelihood** or **least-squares** estimates.

The afore-mentioned conditions for **stationarity** can also be specified through the **characteristic equation**

$$b^p - \phi_{p,1}b^{p-1} - \phi_{p,2}b^{p-2} - \dots - \phi_{p,p} = 0$$

where b is a dummy variable, by stating that the roots of this equation (which means the values of b which satisfy it) should lie inside the unit circle given by $b^2 = 1$.

Accordingly, if **characteristic equation** is written as

$$\prod_{i=1}^p (b - \pi_i) = 0$$

then $|\pi_i| < 1$ for $i = 1, 2, \dots, p$.

Linear autoregressive models

Variance of independent variables

By squaring both sides of the model equation and taking expectations it follows that, because each of the expected values $E[\eta_t \xi_{t-1}]$, $E[\eta_t \xi_{t-2}]$, $\dots$, $E[\eta_t \xi_{t-p}]$ are equal to zero, so

$$1 = \phi_{p,1}\rho_1 + \phi_{p,2}\rho_2 + \phi_{p,3}\rho_3 + \dots + \phi_{p,p}\rho_p + \sigma_\eta^2$$

Hence, σ_η^2 , which is the variance of the independent variables η_t is given by

$$\sigma_\eta^2 = 1 - \phi_{p,1}\rho_1 - \phi_{p,2}\rho_2 - \phi_{p,3}\rho_3 - \dots - \phi_{p,p}\rho_p = 1 - R^2 \quad (3)$$

where $R^2 = \phi_{p,1}\rho_1 + \phi_{p,2}\rho_2 + \phi_{p,3}\rho_3 + \dots + \phi_{p,p}\rho_p$ and is called the **coefficient of determination** or the **square of the multiple correlation coefficient**. R^2 is a **measure of the predictability** of a time series.

Linear autoregressive models

First-order model

The **first-order autoregressive model** $AR(1)$ is known familiarly as a **Markov model** (**first-order Markov chain**) and is given by

$$\xi_t = \phi_{1,1}\xi_{t-1} + \eta_t$$

If this equation is multiplied throughout by ξ_{t-1} and if expectations are taken, because $E[\eta_t\xi_{t-1}] = 0$ in addition to the assumptions $E[\xi_t]^2 = 1$ and $E[\xi_t] = 0$, it follows that

$$\phi_{1,1} = \rho_1$$

where ρ_1 is the **lag-one autocorrelation coefficient** where $-1 < \rho_1 < 1$. From the equation 3, the **variance of the independent variable** η_t is given by

$$\sigma_\eta^2 = 1 - \rho_1^2$$

Again if the equation is multiplied in turn by ξ_{t-1} , ξ_{t-2} , ξ_{t-3} , $\dots$ and ξ_{t+1} , ξ_{t+2} , ξ_{t+3} , $\dots$

$$\rho_k = \rho_1^{|k|} \quad k = 0, \pm 1, \pm 2, \pm 3$$

In a physical sense, this is suggestive of **base flow recession** behaviour (from the ground water component of river flows)

Linear autoregressive models

Second-order model

The **second-order autoregressive model AR(2)** is given by

$$\xi_t = \phi_{2,1}\xi_{t-1} + \phi_{2,2}\xi_{t-2} + \eta_t$$

From the **Yule-Walker** equation, we have that

$$\rho_1 = \phi_{2,1} + \phi_{2,2}\rho_1 \quad \rho_2 = \phi_{2,1}\rho_1 + \phi_{2,2}$$

As explained before, estimates $\hat{\phi}_{2,1}$ and $\hat{\phi}_{2,2}$ of the parameters are obtained by substituting the estimated **serial correlation coefficients** r_1 and r_2 for ρ_1 and ρ_2 . Hence

$$\hat{\phi}_{2,2} = \frac{(r_2 - r_1^2)}{(1 - r_1^2)} \quad (4)$$

$$\hat{\phi}_{2,1} = r_1 - \hat{\phi}_{2,2}r_1 = \frac{r_1(1 - r_2)}{(1 - r_1^2)} \quad (5)$$

The **stationarity** constraints on $\phi_{2,1}$ and $\phi_{2,2}$ are determined as follows. Because

$$\pi_1 = \frac{[\phi_{2,1} - (\phi_{2,1}^2 + 4\phi_{2,2})^{1/2}]}{2} \quad \pi_2 = \frac{[\phi_{2,1} + (\phi_{2,1}^2 + 4\phi_{2,2})^{1/2}]}{2}$$

and $|\pi_1| < 1$ and $|\pi_2| < 1$, it follows that $-1 < \phi_{2,2} < 1$, $\phi_{2,1} + \phi_{2,2} < 1$ and $\phi_{2,2} - \phi_{2,1} < 1$. It can also be shown that for **stationary** conditions that $\rho_1^2 < (\rho_2 + 1)/2$ and $-1 < \rho_1, \rho_2 < 1$.

Linear autoregressive models

General recursive formulate

The equations 4 and 5

$$\hat{\phi}_{2,2} = \frac{r_2 - \hat{\phi}_{1,1}r_1}{1 - \hat{\phi}_{1,1}r_1} \quad \hat{\phi}_{2,1} = \hat{\phi}_{1,1} - \hat{\phi}_{2,2}\hat{\phi}_{1,1}$$

Proceeding in a similar manner from the second- to the third- and so on to higher-order models, we can obtain the estimates of a general **pth-order model** recursively through a pivotal reduction of equation 2. The formulate are

$$\hat{\phi}_{p,p} = \frac{\left(r_p - \sum_{j=1}^{p-1} \hat{\phi}_{p-1,j}r_{p-j}\right)}{\left(1 - \sum_{j=1}^{p-1} \hat{\phi}_{p-1,j}r_j\right)} \quad (6)$$

$$\hat{\phi}_{p,j} = \hat{\phi}_{p-1,j} - \hat{\phi}_{p,p}\hat{\phi}_{p-1,p-j} \quad j = 1, 2, 3, \dots, p-1 \quad (7)$$

Note that this method may not be gives the best estimates. Alternatively, **least-squares** estimates that minimise

$$\sum_{t=1}^N \left(\xi_t - \sum_{i=1}^p \phi_{p,i} \xi_{t-i} \right)^2 \quad (8)$$

where ξ_t , $t = 1, 2, \dots, N$ is an observed sequence and $\xi_{t-i} = 0$ if $t-i < 1$ can be found.

Partial autocorrelations

- ▶ The set of parameters $\phi_{1,1}, \phi_{2,2}, \phi_{3,3}, \dots$, which are the **last coefficients of autoregressive models** of order (p) 1,2,3,..., respectively constitute the so called **partial autocorrelation function (PAF)**.
- ▶ In general, the **partial autocorrelation** $\phi_{p,p}$ is the autocorrelation remaining in the series after fitting a model of order $p - 1$ and removing the linear dependence.
- ▶ Indeed, both the **order** and **type of model** are indicated by the **PAF**.
- ▶ Returning to the particular case of **autoregressive models**, it is important to note that, whereas the **autocorrelation function decrease** to zero only at an infinity lag, in an **autoregressive process of order p** the **partial autocorrelations** $\phi_{k,k}$ are theoretically zero for $k > p$.
- ▶ Therefore, the sample **PAF** is an important tool in determining the order of the model if the **serial correlation function** suggests that the process could be approximated by a **linear autoregressive model**.
- ▶ The **PAF** is also useful for **identifying** other types of processes and as a **diagnostic** tool.

For an **autoregressive process** of order p , the variance of the estimated **partial autocorrelations** is given by

$$\text{Var}[\phi_{k,k}] \approx \frac{1}{N}, \quad k > p$$

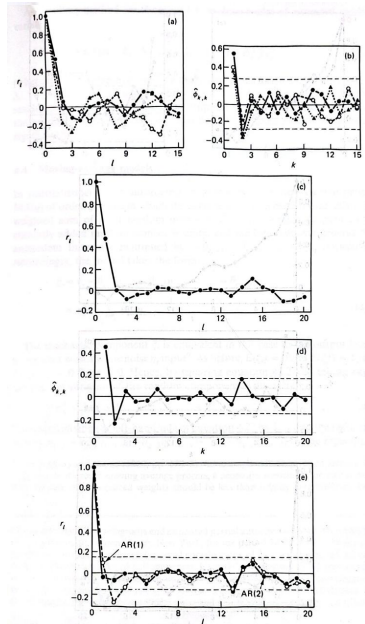
where N is the sample length. A test procedure could be devised by demarcating, say, 95% **confidence limits** which are $1.96/\sqrt{N}$ above and below the horizontal axis of the sample **PAF**. This is on the assumption that the **distribution** of a **partial serial correlation coefficient** with zero expectation is approximately **normal** (despite the distribution may be significantly different from **normal** if $N < 30$). If the parent **time series** is **autoregressive** in form, then the order p of the model would be indicated on the plot when the $\hat{\phi}_{k,k}$ values for $k > p$ are not significantly different from zero.

Partial autocorrelations

Serial correlograms and estimated partial autocorrelograms for an annual streamflow time series with $N = 150$. Note that the broken lines denote 95% confidence limits. a) serial correlograms of three periods of 50 years, b) estimated partial autocorrelograms of three periods of 50 years, c) serial correlogram of 150 years sequence, d) estimated partial autocorrelogram of 150 years sequence and e) serial correlograms of estimated independent residuals $\tilde{\eta}_t$ of 150 year sequence in which the AR(1) and AR(2) models used are given by $\xi_t = 0.0458\xi_{t-1} + \eta_t$ and $\xi_t = 0.582\xi_{t-1} - 0.271\xi_{t-2} + \eta_t$, respectively. An ancillary method for choosing k is to draw a plot of estimated residual variance

$$s_{\eta}^2 \approx \frac{N - k}{N - 2k - 1} \prod_{i=1}^k (t - \hat{\phi}_{i,i}^2) s_{\xi}^2$$

(where s_{ξ}^2 is the variance of the stochastic component) for model lengths $k = 1, 2, 3, \dots$. Accordingly, a suitable order $k = p$ of a model is the order for which the residual variance is minimum, or more appropriately (in practical situations) if the residual variance for models of order greater than p does not decrease significantly.



Moving average models

In contrasting with **autoregressive processes**, a **moving-average process MA(q)** of order q is one in which the current value of a random variable is the weighted sum of $q + 1$ **random numbers**. Here, the weight assigned to the currently added **random number** is unity, and **random numbers** generated at antecedent times are multiplied by $-\theta_{q,1}, -\theta_{q,2}, \dots, -\theta_{q,q}$, respectively. Accordingly, the model takes the form

$$\xi = \eta_t - \theta_{q,1}\eta_{t-1} - \theta_{q,2}\eta_{t-2} - \dots - \theta_{q,q}\eta_{t-q} = \eta_t - \sum_{j=1}^q \theta_{q,j}\eta_{t-j} \quad (9)$$

The stochastic component ξ_t is equivalent in this case to the output from **linear filter** with a **white noise** η_t input. As before, $E[\xi_t] = 0$ and $E[\xi_t^2] = 1$; also, $E[\eta_t\eta_{t-k}] = 0$ for $k \neq 0$. Hence, by squaring the equation 9 and taking expectations, the variance of the random component is given by

$$\sigma_\eta^2 = \frac{1}{1 + \theta_{q,1}^2 + \theta_{q,2}^2 + \dots + \theta_{q,q}^2} \quad (10)$$

Then multiplying the left-hand side of equation 9 by ξ_{t-k} and the right-hand side by $\eta_{t-k} - \theta_{q,1}\eta_{t-1-k} - \theta_{q,2}\eta_{t-2-k} - \dots - \theta_{q,q}\eta_{t-q-k}$, taking expectations and substituting for σ_η^2 from equation 10, we obtain

$$\rho_k = \frac{-\theta_{q,k} + \theta_{q,1}\theta_{q,k+1} + \theta_{q,2}\theta_{q,k+2} + \dots + \theta_{q,q-k}\theta_{q,q}}{1 + \theta_{q,1}^2 + \theta_{q,2}^2 + \dots + \theta_{q,q}^2}, \quad k = 1, 2, 3, \dots, q \quad (11)$$

and $\rho_k = 0$, $k > q$, where ρ_k is the lag- k **autocorrelation coefficient**.

Moving average models

For the first-order moving-average model MA(1), therefore

$$\rho_1 = \frac{-\theta_{1,1}}{1 + \theta_{1,1}^2} \quad (12)$$

Hence, $\theta_{1,1}$ can be estimated from the roots of the equation

$$\hat{\theta}_{1,1}^2 + \frac{\hat{\theta}_{1,1}}{r_1} + 1 = 0 \quad (13)$$

It is seen that one root is the reciprocal of the other and the root which satisfies the invertibility constraint $|\hat{\theta}_{1,1}| > 1$ is the initial estimate of the moving-average parameter. However, such an estimate is found to be inefficient, this means that, for some other value of $\theta_{1,1}$, generated data could look more like the historical data. The suggested procedure is to estimate $\theta_{1,1}$ by minimizing the sum of squares of the residuals which is $\sum_{t=1}^N \tilde{\varepsilon}_t^2$, where $\tilde{\varepsilon}_1 = \xi_1$, $\tilde{\varepsilon}_2 = \xi_2 + \theta_{1,1}\tilde{\varepsilon}_1$, $\tilde{\varepsilon}_3 = \xi_3 + \theta_{1,1}\tilde{\varepsilon}_2$ and ξ_t , $t = 1, 2, 3, \dots, N$, is an observed sequence. The alternative ML procedure is much more difficult for estimation.

Box-Jenkins models: formulation and identification

Generalities

The models described so far are special types of the so-called **Box-Jenkins class of models**. Of these, the **ARMA** model, which includes the **autoregressive**, the **moving-average** and the mixed types, is commonly applied to **stationary** time series. A **backshift** operator B is used in defining the models, so that by using the previous notation $B\xi_t = \xi_{t-1}$, $B^2\xi_t = \xi_{t-2}$ and in general $B^p\xi_t = \xi_{t-p}$. Likewise, $B^q\eta_t = \eta_{t-q}$. There is also a difference operator ∇ such that $\nabla\xi_t = (1 - B)\xi_t = \xi_t - \xi_{t-1}$, $\nabla^2\xi_t = (1 - B)^2\xi_t = (\xi_t - \xi_{t-1}) - (\xi_{t-1} - \xi_{t-2}) = \xi_t - 2\xi_{t-1} + \xi_{t-2}$ and in general $\nabla^d\xi_t = (1 - B)^d\xi_t$. This operator is useful in transforming a **non-stationary** series to one that is **stationary**.

With this notation, **AR(1)** and **MA(1)** models are $(1 - \phi_{1,1}B)\xi_t = \eta_t$ and $\xi_t = (1 - \theta_{1,1}B)\eta_t$, respectively. Because these are equivalent to $\xi_t = (1 - \phi_{1,1}B)^{-1}\eta_t = (1 - \theta_{\infty,1} - \theta_{\infty,2} - \dots)\eta_t$ and $(1 - \theta_{1,1}B)^{-1}\xi_t = (1 - \phi_{\infty,1} - \phi_{\infty,2} - \dots)\xi_t = \eta_t$, respectively, it follows that $AR(1) \equiv MA(\infty)$ and $MA(1) \equiv AR(\infty)$.

The next important requirement is a methodology for identifying the type of model which adequately represents an observed sequence. This means that we should be able to make the best possible choice from the **autoregressive**, **moving-average** and **mixed** models.

Box-Jenkins models: formulation and identification

Autoregressive moving-average (ARMA(p,q)) model

The mixed **autoregressive moving-average (ARMA(p,q)) model** is specified by combining equations 1 and 9

$$\xi_t = \phi_{p,1}\xi_{t-1} + \phi_{p,2}\xi_{t-2} + \dots + \phi_{p,p}\xi_{t-p} + \eta_t - \theta_{q,1}\eta_{t-1} - \theta_{q,2}\eta_{t-2} - \dots - \theta_{q,q}\eta_{t-q} \quad (14)$$

For instance the effect of **linear aquifers** and **independent rainfall** amounts justifies the approximate representation of **river flow processes** through **ARMA** models. They are also analogous to those **conceptual models** in **parametric hydrology** that are based on **linear reservoirs**. The equation 14 can be written as

$$\phi_p(B)\xi_t = \theta_q(B)\eta_t \quad (15)$$

where

$$\phi_p(B) = 1 - \phi_{p,1}B - \phi_{p,2}B^2 - \dots - \phi_{p,p}B^p \quad (16)$$

and

$$\theta_q(B) = 1 - \theta_{q,1}B - \theta_{q,2}B^2 - \dots - \theta_{q,q}B^q \quad (17)$$

Equations 16 and 17 represent two polynomials of order p and q respectively. The prerequisites for **stationarity** and **invertibility** respectively are that the roots of the equations $\phi_p(B) = 0$ and $\theta_q(B) = 0$ or that these roots, which are also termed the zeros of B , should lie **outside** the **unit circle** (these conditions are applicable to **AR(p)** and **MA(q)** processes respectively). Note that the estimation of the parameters of the **ARMA(p,q)** mode is not a straightforward procedure. **Parsimony** must be followed in order to get the minimum $p + q$ parameters.

Box-Jenkins models: formulation and identification

Autoregressive moving-average (ARMA(1,1)) model

In the class of **mixed models**, the simplest is the **ARMA(1,1)** model

$$\xi_t = \phi_{1,1}\xi_{t-1} + \eta_t - \theta_{1,1}\eta_{t-1} \quad (18)$$

Let $E[\eta_t] = E[\xi_t] = 0$, $E[\xi_t^2] = 1$, $\rho_k = E[\xi_t\xi_{t-k}]$ and $E[\xi_t\eta_{t-1}] = \rho_{\xi,\eta}(-1)$. As noted before, $E[\eta_t\eta_{t-k}] = 0$ for $k \neq 0$ and $E[\eta_t\xi_{t-k}] = 0$ for any integer $k > 0$. Initially, if equation 18 is multiplied by ξ_t and expectations are taken, it follows that

$$\begin{aligned} 1 &= \phi_{1,1}\rho_1 + E[\eta_t(\phi_{1,1}\xi_{t-1} + \eta_t - \theta_{1,1}\eta_{t-1})] - \theta_{1,1}\rho_{\xi,\eta}(-1) \\ &= \phi_{1,1}\rho_1 + \sigma_\eta^2 - \theta_{1,1}\rho_{\xi,\eta}(-1) \end{aligned} \quad (19)$$

Then, by multiplying equation 18 by η_{t-1} and ξ_{t-1} in turn and taking expectations

$$\rho_{\xi,\eta}(-1) = (\phi_{1,1} - \theta_{1,1})\sigma_\eta^2 \quad (20)$$

and

$$\rho_1 = \phi_{1,1} - \theta_{1,1}\sigma_\eta^2 \quad (21)$$

Hence

$$\sigma_\eta^2 = \frac{1 - \phi_{1,1}^2}{1 + \theta_{1,1}^2 - 2\phi_{1,1}\theta_{1,1}} \quad (22)$$

and

$$\rho_1 = \frac{(1 - \phi_{1,1}\theta_{1,1})(\phi_{1,1} - \theta_{1,1})}{1 + \theta_{1,1}^2 - 2\phi_{1,1}\theta_{1,1}} \quad (23)$$

Autoregressive moving-average (ARMA(1,1)) model

Also, by multiplying equation 18 by ξ_{t-k} , where $k \geq 2$, and by taking expectations

$$\rho_k = \phi_{1,1}\rho_{k-1} \quad (24)$$

An initial estimate $\hat{\phi}_{1,1}$ of $\phi_{1,1}$ is obtained from r_2/r_1 (where r_1 and r_2 are the lag-one and lag-two serial correlation coefficients) by using equation 24. The substitution of $\hat{\phi}_{1,1}$ and r_1 in equation 23 gives an initial estimate $\hat{\theta}_{1,1}$ of $\theta_{1,1}$. Then, given the **stationarity** and **invertibility** constraints $-1 < \phi_{1,1}, \theta_{1,1} < 1$ a **grid search** is made, and the two parameters are estimated by minimising the sum of squares of residuals which is $\sum_{t=1}^N \tilde{\varepsilon}_t^2$, where $\tilde{\varepsilon}_1 = \tilde{\xi}_1$, $\tilde{\varepsilon}_2 = \tilde{\xi}_2 - \phi_{1,1}\tilde{\xi}_1 + \theta_{1,1}\tilde{\varepsilon}_1$, $\tilde{\varepsilon}_3 = \tilde{\xi}_3 - \phi_{1,1}\tilde{\xi}_2 + \theta_{1,1}\tilde{\varepsilon}_2$, $\dots$, and $\tilde{\xi}_t$, $t = 1, 2, 3, \dots, N$ is an observed sequence.

Box-Jenkins models: formulation and identification

Autoregressive integrated moving-average (ARIMA(p,d,q)) model

The **ARIMA** model is a generalized version of the **ARMA** model which can account for non-stationarity behaviour by means of a d th-order difference operator ∇^d . By using the notation explained above it can be represented by

$$\phi_p(B)\nabla^d\xi_t = \theta_q(B)\eta_t \quad (25)$$

By changing the variable ξ_t , this can be transformed to an **ARMA(p,q)** type. For example, if $\psi_t = \nabla^d\xi_t$,

$$\phi_p(B)\psi_t = \theta_q(B)\eta_t \quad (26)$$

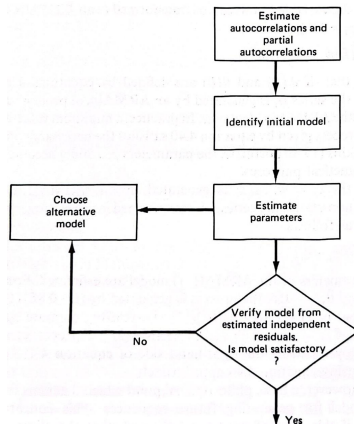
This means that, if $\phi(B)$ and $\theta(B)$ are defined by equations 16 and 17, respectively, the series ψ_t is generated by an **ARMA(p,q)** process; correspondingly, ξ_t is of the **ARIMA(p,d,q)** type. In practice, a **minimum value** of d is found so that the process given by equation 26 satisfies the necessary constraints. As Box and Jenkins postulate, the parameters p , d and q need not be greater than 2 for practical purposes.

Consider the case when ξ_t is generated by an **ARIMA(1,1,1)** process. Initially, the non-stationary series ξ_t is transformed into a stationary series ψ_t by differencing as $\psi_t = \xi_t - \xi_{t-1}$. Then the parameters of **ARMA(1,1)** model are estimated. An **ARIMA(2,2,2)** process can be similarly investigated by replacing $\psi_t = \xi_t - 2\xi_{t-1} + \xi_{t-2}$ in equation 26.

Box-Jenkins models: formulation and identification

Model identification and estimation

- ▶ An iterative procedure for fitting **Box-Jenkins** model is given in the figure. This procedure is performed recursively, where the essential steps are **identification**, **estimation** and **verification** of the model **type**, **order** and **parameters**.
- ▶ The **serial correlogram** (ρ_i) is the initial requirement. Also, the estimation of the **partial autocorrelation** ($\phi_{i,i}$) is equally important.
- ▶ Note that complications arise from unremoved non-stationary effects. This can be identified in the **serial correlogram** and in the **partial autocorrelation**.
- ▶ The behaviour of **acf** and **pcf** functions are different. For instance, in the **AR(1)** model, the **acf** decrease exponentially, whereas the **pcf** are theoretically zero except for $\phi_{1,1} \neq 0$. The situation is the opposite for a **MA(1)** model.
- ▶ In an **ARMA(1,1)** process, both functions decay exponentially after $k = 1$. For identifying these and other types, see following table.



Box-Jenkins models: formulation and identification

Model identification and estimation

<i>Model type</i>	<i>Autocorrelation function</i>	<i>Partial autocorrelation function</i>
AR(1), first-order autoregressive (Markov)	Exponential decrease	$\phi_{1,1} \neq 0$ $\phi_{i,i} = 0$ for $i = 2, 3, 4, \dots$
AR(p), p th-order autoregressive	Mixed type of damping from lag 1	$\phi_{i,i} \neq 0$ for $i \leq p$ $\phi_{i,i} = 0$ for $i > p$
MA(1), first-order moving-average	$\rho_1 \neq 0$ $\rho_i = 0$ for $i = 2, 3, 4, \dots$	Exponential decrease
MA(q), q th-order moving-average	$\rho_i = 0$ for $i > q$ $\rho_i \neq 0$ for $i \leq q$	Mixed type of damping from lag 1
ARMA(1, 1), autoregressive moving-average	Exponential decrease after lag 1	Exponential decrease after lag 1
ARMA(p, q), autoregressive moving-average	Mixed type of damping after lag $q + 1$	Mixed type of damping after lag $p - q$

Box-Jenkins models: formulation and identification

Model identification and estimation

1. The shapes of the sample **acf** and **pcf** functions are examined for practical identification purposes.
2. If the indications are that the process is an **autoregressive** type, a **significance test** is made to determine its order. This is based on the assumption that the estimated **pcf** function $\hat{\phi}_{i,i}$, $i > p$, of an **AR(p)** process are normally distributed with zero mean and variance $1/N$, where N is the number of observations. Similarly, for an **MA(q)** process, the **serial correlation** r_l , $l > q$, are taken to be **normally distributed** with zero mean and variance

$$\text{Var}(r_l) \approx \frac{(1 + 2 \sum_{k=1}^q r_k^2)}{N} \quad l > q \quad (27)$$

3. Estimate the parameters of the model according to the procedure explained before.
4. After **identification** and **estimation**, **diagnostic checking** or **verification** is made by testing the residuals thus obtained for independence. For general **ARMA(p,q)** model described as

$$\phi(B)\xi_t = \theta(B)\eta_t$$

the estimated residuals are, therefore $\hat{\eta}_t = \hat{\theta}^{-1}(B)\hat{\phi}(B)\hat{\xi}_t$, where $\hat{\theta}$ and $\hat{\phi}$ are polynomials (in B) in which the coefficients are estimated parameters, and $\hat{\xi}_t$, $t = 1, 2, 3, \dots, N$, is an observed sequence.

Box-Jenkins models: formulation and identification

Model identification and estimation

1. To clarify this, consider a **pth-order autoregressive model** $\phi(B)\xi_t = \eta_t$ given explicitly by equation 1. In this case, the **residuals** are estimated as follow

$$\tilde{\eta}_t = \tilde{\xi}_t - \sum_{k=1}^p \hat{\phi}_{p,k} \tilde{\xi}_{t-k} \quad (28)$$

where $\tilde{\xi}_{t-k} = 0$, if $t - k < 1$. Like wise for the **moving-average model** given by equation 9, estimated **residuals** are found from

$$\tilde{\eta}_t = \tilde{\xi}_t + \sum_{k=1}^q \hat{\theta}_{q,k} \tilde{\eta}_{t-k} \quad (29)$$

where $\tilde{\eta}_{t-k} = 0$, if $t - k < 1$. Similarly, the estimated **residuals** obtained by applying the **ARMA(p,q)** model are given by

$$\tilde{\eta}_t = \tilde{\xi}_t - \sum_{k=1}^p \hat{\phi}_{p,k} \tilde{\xi}_{t-k} + \sum_{k=1}^q \hat{\theta}_{q,k} \tilde{\eta}_{t-k} \quad (30)$$

where $\tilde{\xi}_{t-k} = \tilde{\eta}_{t-k} = 0$, if $t - k < 1$.

Box-Jenkins models: formulation and identification

Model identification and estimation

- ▶ If the **ARMA(p,q)** family of models do not fit the observed sequence, the (non-stationary) **ARIMA(p,d,q)** types may be tried. Accordingly, d th-order differencing is carried out on the observed ξ_t sequence, as explained in before to form the sequence $\tilde{\psi}_t = \nabla^d \xi_t$, $t = 1, 2, 3, \dots$. Here, a minimum value of d is chosen by trial and error so that a model from the **ARMA(p,q)** family can be fitted to the $\tilde{\psi}_t$ sequence, as previously discussed.
- ▶ For the purpose of graphical verification, **serial correlations** r_l of the estimated residuals $\tilde{\eta}_t$ are obtained after fitting the appropriate model from those represented by equations 28, 29 and 30. The **confidence limits** are drawn on the **serial correlogram** so obtained to test the hypothesis that the $\tilde{\eta}_t$ values have an independent **normal distribution**, in which case the r_l values are approximately **normally distributed** with zero mean and variance approximately equal to $1/N$, where N is the number of data.